described in terms of a *nuclear cross-section.* Let us assume that nuclei are spheres, neutrons are points, and no force of any kind is acting between them at all. In that case, the larger are the cross-sections of the spheres, the greater would be the probability of striking a nucleus with neutrons shot at it at random. In this model, the nuclei would apparently have cross-sections of different sizes for neutrons of different velocities. Under particularly favourable conditions, this nuclear cross-section may be about 10,000 times larger than the actual geometrical cross-sectional area of the nucleus. This means an exceedingly high probability of capture. This is why Fermi, who discovered this circumstance, first applied the device already mentioned of slowing down the neutrons (which are always rather fast when they are first produced) to thermal velocity. He made them pass through a hydrogen-containing substances, such as water or paraffin. Hydrogen is the best substance for this purpose, because the mass of a proton is approximately the same as that of a neutron, and also because the conditions for a rapid exchange of energy are most favourable here, in conformity with the laws of elastic impacts. When such slow neutrons are captured by a nucleus, they are accelerated within the range of attraction of the nuclear forces sufficiently to produce that heating of the nucleus which has already been discussed.

Conversely, too, it is naturally easier for a neutron to emerge out of such a heated nucleus than it would be for a charged particle. In fact, the charged particle has to ‘climb over’ the potential barrier in order to get out, whereas for the neutron the potential barrier is non-existent. This is the reason why nuclear transmutations involving the emission of a proton or an alpha particle are relatively rare in heavy nuclei where the potential barrier is high.

Of course, when a nucleus is heated to a very high degree, a larger number of charged and uncharged particles can come out of it, analogously to the evaporation of a liquid drop. With laboratory apparatus it is very difficult to impart such high energies to the nucleus. But particles possessing the vast energy of 1,000 Mev., or even more, are present in cosmic radiation. If such a particle happens to hit a nucleus, the result of the encounter is that the nucleus is heated to such a high degree and emits numerous protons and neutrons, at times even heavier fragments, such as helium or lithium nuclei. When a nuclear disintegration of this kind actually takes place in the emulsion on a photographic plate, the nuclear particles leave tracks which become visible when the plate is developed. This technique was developed by Blau, Wambacher and Schopper, and has recently been improved considerably by Powell and his collaborators. Figure 25A shows such a photograph. It is a much enlarged reproduction; the range of the protons in the emulsion is actually less than 1 mm. We are thus looking here at an actual record of the disintegration of an atom, in which at least forty particles or so have ‘evaporated’ out of the nucleus. For, in addition to the charged particles, many